

## 2-Hour Technical Assessment Documentation

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1. What is the correct use of using statement?

- A. Improves syntax
- B. Handles exceptions
- C. Automatically releases unmanaged resources ✓
- D. Improves execution speed

2. Which type is immutable in C#?

- A. StringBuilder
- B. Array
- C. string ✓
- D. List

3. What is boxing in C#?

- A. Reference to reference
- B. Value to reference type ✓
- C. Reference to value
- D. Object cloning

4. Which collection is fastest for lookup?

- A. List
- B. Dictionary ✓
- C. ArrayList
- D. Queue

5. What does async keyword do?

- A. Creates thread
- B. Blocks thread
- C. Enables asynchronous execution ✓
- D. Stops execution

6. Which access modifier is most restrictive?

- A. public
- B. internal
- C. protected
- D. private ✓

7. What is CLR?

- A. Compiler
- B. Runtime environment ✓
- C. Garbage collector
- D. Debugger

8. Which keyword is used for inheritance?

- A. this
- B. base
- C. inherit
- D. : ✓

9. Which is reference type?

- A. int
- B. struct
- C. enum
- D. class ✓

10. What is garbage collection?

- A. Manual memory cleanup
- B. Automatic memory management ✓
- C. File deletion
- D. Object creation

11. What is LINQ mainly used for?

- A. File handling
- B. Data querying ✓
- C. Threading
- D. Serialization

12. Which block is used to handle exceptions?

- A. try-catch ✓
- B. if-else
- C. for-loop
- D. switch

13. What is a delegate?

- A. Data type
- B. Reference to method ✓
- C. Class
- D. Namespace

14. Interface contains?

- A. Fields
- B. Constructors
- C. Abstract methods only ✓
- D. Implementations

15. Abstract class can have?

- A. No methods
- B. Only abstract methods
- C. Abstract + non-abstract methods ✓
- D. Static methods only

16. Value types are stored in?

- A. Heap
- B. Stack ✓
- C. Cache
- D. Register

17. Reference types are stored in?

- A. Stack
- B. Heap ✓
- C. CPU
- D. ROM

18. IDisposable is used for?

- A. Memory allocation
- B. Resource cleanup ✓
- C. Thread safety
- D. Logging

19. Task is better than Thread because?

- A. Slower
- B. Uses more memory
- C. Lightweight & managed by CLR ✓
- D. Blocks execution

20. Struct is?

- A. Reference type
- B. Value type ✓
- C. Interface
- D. Delegate

21. Sealed class means?

- A. Can be inherited
- B. Cannot be inherited ✓
- C. Abstract
- D. Partial

22. Static constructor is used for?

- A. Object creation
- B. Initializing static data ✓
- C. Garbage collection
- D. Inheritance

23. Partial class allows?

- A. Multiple inheritance
- B. Splitting class in files ✓
- C. Overriding methods
- D. Hiding members

24. Nullable type is represented by?

- A. !
- B. ? ✓
- C. \*
- D. &

25. Which keyword frees unmanaged resources?

- A. dispose
- B. free
- C. using ✓
- D. delete

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26. JavaScript is \_\_\_ typed language. is \_\_\_ typed language.

- A. Strongly
- B. Statically
- C. Dynamically ✓
- D. Compiled

27. Which keyword declares block-scoped variable?

- A. var
- B. let ✓
- C. const
- D. static

28. What is closure?

- A. Loop
- B. Function inside function with scope access ✓
- C. Event
- D. Callback

29. Which is asynchronous?

- A. for loop
- B. if condition
- C. setTimeout ✓
- D. switch

30. Which operator checks value and type?

- A. ==
- B. =
- C. === ✓
- D. !==

31. What is hoisting in JavaScript?

- A. Moving functions to bottom
  - B. Moving variables & functions to top of scope ✓
  - C. Executing code faster
  - D. Memory allocation
- 

32. Which keyword is hoisted completely?

- A. let
  - B. const
  - C. var ✓
  - D. static
- 

33. What is event bubbling?

- A. Event goes from child to parent ✓
  - B. Event goes from parent to child
  - C. Event stops immediately
  - D. Event is asynchronous
-

34. How to stop event bubbling?

- A. stop()
  - B. prevent()
  - C. stopPropagation() ✓
  - D. cancel()
- 

35. Callback function is:

- A. Function called automatically by browser
  - B. Function passed as argument to another function ✓
  - C. Function returning function
  - D. Function with delay
- 

36. Main problem with callbacks?

- A. Slow execution
  - B. Callback hell (nested callbacks) ✓
  - C. Memory leak
  - D. Compilation error
- 

37. Promise has which states?

- A. start, stop
  - B. open, close
  - C. pending, fulfilled, rejected ✓
  - D. success, failure
- 

38. Which keyword handles asynchronous code easily?

- A. await
  - B. async/await ✓
  - C. promise
  - D. then
-

39. async function always returns?

- A. Value
  - B. Boolean
  - C. Promise ✓
  - D. Object
- 

40. JSON.parse() is used to:

- A. Convert object to string
  - B. Convert JSON string to object ✓
  - C. Validate JSON
  - D. Store JSON
- 

41. JSON.stringify() is used to:

- A. Convert object to JSON string ✓
  - B. Convert JSON to XML
  - C. Parse JSON
  - D. Compress JSON
- 

42. DOM stands for?

- A. Document Object Model ✓
  - B. Data Object Method
  - C. Digital Order Model
  - D. Document Oriented Map
- 

43. Arrow functions were introduced in?

- A. ES3
  - B. ES5
  - C. ES6 ✓
  - D. ES7
-

44. Spread operator is represented by?

- A. \*\*
  - B. %%
  - C. ... ✓
  - D. =>
- 

45. Difference between LocalStorage & SessionStorage?

- A. Speed
  - B. Size
  - C. Persistence (SessionStorage clears on tab close) ✓
  - D. Syntax
- 

46. jQuery is a \_\_\_\_ library.

- A. Server-side
- B. JavaScript ✓
- C. CSS
- D. Database

47. Which symbol is used for jQuery?

- A. #
- B. &
- C. %
- D. \$ ✓

48. `$(document).ready()` executes when?

- A. Page refresh
- B. DOM loaded ✓
- C. Server responds
- D. CSS loads

49. Which method performs POST?

- A. \$.get
- B. \$.load
- C. \$.post ✓
- D. \$.fetch



50. jQuery selector is used to?

- A. Create variables
  - B. Select HTML elements ✓
  - C. Compile code
  - D. Connect database
- 

51. Which selector selects element by ID?

- A. .id
  - B. #id ✓
  - C. id#
  - D. \$id
- 

52. Which selector selects elements by class?

- A. #class
  - B. .class ✓
  - C. class#
  - D. \*class
- 

53. Which event fires when button is clicked?

- A. hover()
  - B. mouseover()
  - C. click() ✓
  - D. press()
- 

54. What is event handling in jQuery?

- A. Styling elements
  - B. Handling user actions like click, keypress ✓
  - C. Database call
  - D. Page reload
-

55. Method chaining in jQuery means?

- A. Writing multiple scripts
  - B. Calling multiple methods on same element in one statement ✓
  - C. Linking files
  - D. Looping methods
- 

56. Which method hides element?

- A. display(false)
  - B. hide() ✓
  - C. invisible()
  - D. remove()
- 

57. Which method toggles visibility?

- A. toggle() ✓
  - B. switch()
  - C. hideShow()
  - D. flip()
- 

58. Which method is used to get/set attribute value?

- A. prop()
  - B. val()
  - C. attr() ✓
  - D. html()
- 

59. Which method gets/sets form control value?

- A. attr()
  - B. text()
  - C. val() ✓
  - D. prop()
-

60. Which method removes element from DOM?

- A. delete()
  - B. remove() ✓
  - C. hide()
  - D. detach()
- 

61. AJAX stands for?

- A. Asynchronous JavaScript and XML ✓
- B. Advanced Java XML
- C. Active JS
- D. Async Java

62. Main benefit of AJAX?

- A. Security
- B. Faster UI without reload ✓
- C. Database access
- D. Styling

63. Which HTTP method is secure for data?

- A. GET
- B. POST ✓
- C. PUT
- D. DELETE

64. XMLHttpRequest is used to?

- A. Style page
  - B. Send HTTP requests asynchronously ✓
  - C. Access database
  - D. Handle UI
- 

65. AJAX allows?

- A. Full page reload
  - B. Partial page update without reload ✓
  - C. Only server-side execution
  - D. Only GET request
-

66. Fetch API is?

- A. Database API
  - B. Modern replacement of XMLHttpRequest ✓
  - C. jQuery function
  - D. CSS framework
- 

67. fetch() returns?

- A. JSON
  - B. Promise ✓
  - C. Object
  - D. String
- 

68. Which method converts response to JSON?

- A. JSON.parse()
  - B. response.json() ✓
  - C. fetch.json()
  - D. parseJSON()
- 

69. Success callback is executed when?

- A. Request fails
  - B. Request is sent
  - C. Response is successful ✓
  - D. Page reloads
- 

70. Error callback executes when?

- A. HTTP status is success
  - B. Server returns error or request fails ✓
  - C. Page loads
  - D. Data parsed
-

71. Which status code means success?

- A. 404
  - B. 500
  - C. 200 ✓
  - D. 302
- 

72. Which status code means server error?

- A. 200
  - B. 404
  - C. 500 ✓
  - D. 301
- 

73. Asynchronous request means?

- A. Code waits for response
  - B. Code executes in sequence
  - C. Code continues without waiting for response ✓
  - D. Code stops execution
- 

74. Difference between Sync and Async?

- A. Speed
  - B. Syntax
  - C. Blocking vs Non-blocking execution ✓
  - D. Data type
- 

75. AJAX flow in MVC?

- A. View → Database
- B. View → Controller → Model → Controller → View (JSON/Partial View) ✓
- C. Controller → View → DB
- D. Model → View → Controller

76. ADO.NET is used for?

- A. UI
- B. Database access ✓
- C. Styling
- D. Hosting

77. Which is connected architecture?

- A. DataSet
- B. DataTable
- C. DataReader ✓
- D. DataAdapter

78. Which is disconnected?

- A. SqlCommand
- B. SqlDataReader
- C. DataSet ✓
- D. SqlConnection

79. Fastest way to read data?

- A. DataTable
- B. DataSet
- C. DataReader ✓
- D. Adapter

80. Which prevents SQL Injection?

- A. Inline SQL
- B. Dynamic SQL
- C. Parameters ✓
- D. Loops

81. Stored Procedure is?

- A. Table
- B. View
- C. Precompiled SQL code in DB ✓
- D. Trigger

---

82. Advantage of Stored Procedure?

- A. Faster execution
- B. Security

- C. Reusability
  - D. All of the above ✓
- 

83. ExecuteNonQuery() is used for?

- A. SELECT
  - B. INSERT, UPDATE, DELETE ✓
  - C. Scalar values
  - D. Dataset
- 

84. ExecuteScalar() returns?

- A. Dataset
  - B. DataReader
  - C. Single value (first row, first column) ✓
  - D. Multiple rows
- 

85. ExecuteReader() returns?

- A. Dataset
  - B. DataTable
  - C. SqlDataReader ✓
  - D. Scalar value
- 

86. SqlCommandBuilder is used for?

- A. Creating tables
  - B. Auto-generating INSERT/UPDATE/DELETE commands ✓
  - C. Executing SP
  - D. Managing connection
- 

87. Which architecture uses DataAdapter & DataSet?

- A. Connected
- B. Disconnected ✓

- C. Hybrid
  - D. Static
- 

88. `DataAdapter.Fill()` does what?

- A. Saves data to DB
  - B. Deletes data
  - C. Loads data from DB to `DataSet` ✓
  - D. Opens connection permanently
- 

89. `DataAdapter.Update()` is used to?

- A. Fetch data
  - B. Apply `DataSet` changes to database ✓
  - C. Delete dataset
  - D. Create connection
- 

90. Transaction ensures?

- A. Fast execution
  - B. Security
  - C. ACID properties (Atomicity, Consistency, Isolation, Durability) ✓
  - D. Caching
- 

91. Which class handles transactions?

- A. `SqlTransaction` ✓
  - B. `SqlAdapter`
  - C. `SqlReader`
  - D. `SqlCommandBuilder`
- 

92. Connection pooling means?

- A. Closing connection always
- B. Reusing database connections ✓



- C. Creating multiple DBs
  - D. Caching data
- 

### 93. Benefit of connection pooling?

- A. Security
  - B. Reduced overhead & better performance ✓
  - C. More memory usage
  - D. Slow response
- 

### 94. Using statement is used for?

- A. Exception handling
  - B. Looping
  - C. Automatic resource disposal ✓
  - D. Threading
- 

### 95. Which interface supports using statement?

- A. IEnumerable
  - B. IDisposable ✓
  - C. ICloneable
  - D. IComparable
- 

### 96. CRUD stands for?

- A. Create Read Update Delete ✓
  - B. Copy Remove Update Delete
  - C. Create Replace Use Delete
  - D. Connect Read Update Drop
- 

### 97. CRUD using Stored Procedure improves?

- A. Performance
- B. Security

- C. Maintainability
  - D. All of the above ✓
- 

98. Which object executes Stored Procedures?

- A. SqlConnection
  - B. SqlCommand ✓
  - C. SqlDataAdapter
  - D. DataSet
- 

99. Proper way to handle DB exceptions?

- A. Ignore
  - B. try-catch-finally ✓
  - C. if-else
  - D. goto
- 

100. Best practice for DB code?

- A. Hardcoded SQL
  - B. Open connection forever
  - C. Use SP + using + exception handling ✓
  - D. No error handling
-